
NSSR Summer Research Fellowship

NSRF

Full Program Schedule and Curriculum Guide
3-Month Fellowship • 12 Weeks • Training • Research • Publication

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1. Program Overview

The NSSR Summer Research Fellowship (NSRF) is a structured three-month program designed to guide students from undergraduate to doctoral level through the complete research lifecycle. The program combines foundational academic training, hands-on research practice, scientific writing, and formal presentation into a cohesive experience that builds lasting research competence.

The fellowship unfolds across four phases: a training and proposal phase (Weeks 1–3), a group development and writing skills phase (Weeks 4–5), an active research implementation phase (Weeks 6–11), and a final symposium phase (Week 12).

Program at a Glance

Phase	Weeks	Focus Area	Key Deliverable
Phase 1	1 – 3	Research Foundations, Tools & Individual Proposals	Individual Proposal Presentation + Group Formation
Phase 2	4 – 5	Group Research Development & Writing Training	Group Proposal Presentation + Paper Outline
Phase 3	6 – 11	Active Research Implementation	Weekly Reports + Progress Presentations
Phase 4	12	Final Research Symposium	Final Report + Research Showcase + Certificate

Daily Session Format: Weeks 1 and 2

Each training day in Weeks 1 and 2 is 2.5 hours long, structured consistently as follows:

Segment	Duration	Content	Mode
Core Course Session	2 hours (120 min)	Structured curriculum covering theory, concepts, practical tools, and hands-on activities. Content is tiered for Bachelor, Master, and PhD students.	Interactive Lecture
Research Field Spotlight	30 minutes	Expert-led introduction to a specific research discipline: scope, current frontiers, career entry points, and open research questions.	Talk + Q&A

Phase 1: Week-by-Week Schedule

Week	Day	Course Topic (2 Hours)	Field Spotlight (30 Min)	Presenter
1	Day 1	Introduction to Research and the Scientific Method	Particle Physics	Ash-ray
	Day 2	Scientific Literature, Databases & Source Evaluation	Astrophysics	Sub-ash
	Day 3	Reading and Critically Evaluating a Scientific Paper	Plasma Physics	Manish
2	Day 4	Reference Management, Citations & Academic Integrity	Biophysics	Om
	Day 5	LaTeX, Overleaf & Principles of Scientific Writing	Material Science	Dhirendra
	Day 6	Research Proposal Writing & Presentation Skills	Econophysics	Prasant
3	Days 7–9	Individual Proposal Presentations 8–10 min per student + Q&A	Group Formation + Email Notification	All Mentors

2. Phase 1: Foundation Training (Weeks 1–3)

Phase 1 establishes the intellectual and practical groundwork every researcher needs before beginning independent work. Over six structured training sessions across two weeks, students develop skills in literature navigation, reference management, scientific writing, and proposal formulation. Week 3 concludes the phase with individual proposal presentations and group formation.

Week 1: Research Foundations

Week 1 introduces students to the culture and process of scientific inquiry. By the close of the week, students will understand the research lifecycle, search and evaluate academic literature confidently, and read any scientific paper with structured comprehension.

Day 1: Introduction to Research and the Scientific Method | 2 Hours

Learning Objectives

- Understand what research is and why it is central to scientific and societal progress.
- Distinguish between types of research and articulate which applies to your field.
- Appreciate the role of ethics and integrity as non-negotiable foundations of academic work.

Session Outline

Time	Topic	Activity / Notes
0:00 – 0:20	What is Research? Definition, purpose, and the role of curiosity in science	Opening discussion: what does research mean to each student?
0:20 – 0:45	Types of Research Pure/fundamental, applied, experimental, theoretical, computational	Field examples drawn from physics, biology, economics, and social science.
0:45 – 1:10	The Scientific Method Observation, hypothesis, experiment, analysis, conclusion, and dissemination	Case study walkthrough using a real published paper as an example.
1:10 – 1:35	Research Ethics Plagiarism, data fabrication, authorship norms, conflicts of interest, responsible research conduct	Scenario-based group discussion on ethical dilemmas in science.
1:35 – 2:00	The Research Lifecycle From initial idea to publication: stages, realistic timelines, and expectations	Assignment brief: read one paper, submit a short reflection before Day 2.

Level-Specific Notes

- Bachelor students: Focus on the purpose of research, basic scientific method, and the importance of citing sources.
- Master students: Discuss the difference between literature-based and empirical research; how a thesis relates to a journal article.
- PhD students: Explore research paradigms (positivism, interpretivist, mixed), identifying knowledge gaps, and establishing novelty.

Field Spotlight (30 Min) : Particle Physics | Presenter: Ash-ray

- Overview of the Standard Model: fundamental particles, forces, and how they interact.
- Current frontiers: Large Hadron Collider (LHC) experiments, neutrino oscillations, dark matter searches.
- What research in particle physics looks like in practice: theory, simulation, and large-scale experimentation.
- Q&A: How can students at any level engage with particle physics research from their institution?

Day 2 : Scientific Literature, Databases & Source Evaluation | 2 Hours

Learning Objectives

- Navigate major academic databases and perform targeted searches efficiently.
- Assess source credibility and distinguish reputable journals from predatory publications.
- Understand open-access publishing and how to obtain full-text papers legally.

Session Outline

Time	Topic	Activity / Notes
0:00 – 0:25	Academic Databases Google Scholar, ArXiv, PubMed, Scopus, Web of Science, IEEE Xplore: purpose and differences	Live demo: searching the same topic across three databases and comparing results.
0:25 – 0:50	Effective Search Strategies Boolean operators (AND, OR, NOT), keyword selection, date filters, citation counts	Hands-on: every student searches their area of interest using at least two databases.
0:50 – 1:15	Evaluating Source Quality Impact factor, h-index, quartile rankings (Q1–Q4), predatory journals (Beall's List)	Compare a predatory journal listing against a reputable Q1 journal.
1:15 – 1:40	Open Access and Preprints ArXiv, DOAJ, PubMed Central, institutional repositories, and how to access paywalled content	Practical: students access a full-text paper using legal free methods.

1:40 – 2:00	Building a Literature Map Identifying seminal papers, review articles, citation networks, and research trends	Students compile an initial list of 5 relevant papers in their interest area.
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Level-Specific Notes

- Bachelor: Master basic Google Scholar searching and filtering by date and citation count.
- Master: Conduct a systematic literature search; distinguish a review article from primary research.
- PhD: Map citation networks and trace seminal contributions; identify leading authors and active research groups in a subfield.

Field Spotlight (30 Min) : Astrophysics | Presenter: Sub-ash

- Overview of astrophysics: stellar evolution, galaxies, black holes, dark energy, and observational cosmology.
- Observational vs. theoretical astrophysics: instruments, missions, and modelling approaches.
- Current space missions: James Webb Space Telescope (JWST), Chandra X-ray Observatory, LIGO gravitational-wave detector.
- Research entry points: public astronomical datasets (NASA, ESA), simulation codes, and collaboration opportunities.

Day 3: Reading and Critically Evaluating a Scientific Paper | 2 Hours

Learning Objectives

- Navigate any research paper using the IMRAD structure efficiently.
- Extract key claims, methodology choices, and principal results accurately.
- Apply critical thinking to identify a paper's strengths, limitations, and open questions.

Session Outline

Time	Topic	Activity / Notes
0:00 – 0:20	IMRAD Structure Abstract, Introduction, Methodology, Results, Discussion, Conclusion, References explained	Annotated example paper distributed; key sections highlighted.
0:20 – 0:45	Strategic Reading Order Abstract → Figures & Tables → Introduction → Discussion → Methods (and why)	Discussion: reading in order vs. reading strategically.
0:45 – 1:15	Critical Evaluation Identifying the core research question, assessing methodology rigor, evaluating conclusions vs. evidence	Pair exercise: students critique a selected paper and share observations.

1:15 – 1:40	Key Scientific Terminology p-value, confidence interval, statistical significance, control group, peer review, replication	Glossary handout; students quiz each other on definitions.
1:40 – 2:00	Note-Taking Frameworks Cornell method, annotation techniques, summary grids for managing multiple papers	Template distributed; students annotate a section of their chosen paper.

Level-Specific Notes

- Bachelor: Focus on identifying the main finding and understanding the overall structure.
- Master: Evaluate the methodology, is it reproducible? Are the conclusions justified by the data?
- PhD: Conduct a comparative critical analysis of two papers on similar topics; discuss paradigm shifts and contested findings.

Field Spotlight (30 Min): Plasma Physics | Presenter: Manish

- Plasma as the fourth state of matter and its prevalence throughout the observable universe.
- Core concepts: Debye shielding, magnetohydrodynamics (MHD), plasma confinement and instabilities.
- Applications: nuclear fusion energy (ITER, tokamak reactors), space and astrophysical plasmas, industrial plasma processing.
- Research pathways and key beginner resources textbooks, simulation codes, and open datasets.

Week 2: Research Tools and Proposal Development

Week 2 equips students with the practical tools of modern research: reference management software, LaTeX typesetting, and scientific writing principles. The week culminates in a focused proposal-writing session that prepares every student for their Week 3 individual presentation.

Day 4: Reference Management, Citations & Academic Integrity | 2 Hours

Learning Objectives

- Set up and use Zotero to collect, organize, and cite references.
- Apply APA, IEEE, and MLA citation styles correctly and know when to use each.
- Understand the forms of plagiarism and adopt strategies to maintain academic integrity.

Session Outline

Time	Topic	Activity / Notes
0:00 – 0:30	Zotero Setup and Use Installation, browser connector, adding references, organizing collections, cloud sync	Hands-on: every student installs Zotero and adds 3 papers collected in previous sessions.
0:30 – 0:55	Citation Styles in Depth APA 7th Edition, IEEE, MLA 9th: structure, punctuation, when each is used	Comparative examples: the same source cited in all three styles side by side.
0:55 – 1:20	Citing in Documents Zotero plugin for Word and BibTeX export for LaTeX; generating in-text citations and bibliographies	Live demo: insert citations and auto-generate a bibliography in two formats.
1:20 – 1:45	Plagiarism in Depth Verbatim, mosaic, paraphrasing, and self-plagiarism; detection tools (Turnitin, iThenticate)	Group discussion: borderline case scenarios where is the line?
1:45 – 2:00	Academic Integrity Policy NSSR standards, authorship criteria, citing unpublished work, conference vs. journal norms	Q&A and session summary.

Level-Specific Notes

- Bachelor: Install Zotero, practice consistent APA or IEEE citations, and understand what constitutes plagiarism.
- Master: Use Zotero for large literature collections; integrate it with LaTeX using BibTeX export.
- PhD: Explore custom citation style editing, managing citations for long manuscripts, and journal-specific submission checklists.

Field Spotlight (30 Min): Biophysics | Presenter: Om

- Biophysics: applying the methods and principles of physics to understand biological systems at molecular, cellular, and systems levels.
- Key research topics: protein folding and dynamics, cell membrane mechanics, neural signal propagation, and DNA physics.
- Experimental tools: Atomic Force Microscopy (AFM), optical tweezers, FRET spectroscopy, and patch-clamp electrophysiology.
- Entry points for students: computational biophysics modeling, lab collaborations, and open biological datasets.

Day 5 : LaTeX, Overleaf & Principles of Scientific Writing | 2 Hours**Learning Objectives**

- Create structured scientific documents in Overleaf using LaTeX.
- Typeset equations, insert figures, and format tables correctly.
- Apply core principles of scientific writing: clarity, precision, conciseness, and objectivity.

Session Outline

Time	Topic	Activity / Notes
0:00 – 0:20	Why LaTeX? Advantages for scientific typesetting: equations, cross-references, bibliography automation, journal templates	Side-by-side: Word-produced document vs. LaTeX output of the same content.
0:20 – 0:50	Overleaf Fundamentals Creating a project, document classes (article, IEEE, APA), sections, labels, cross-references	Each student creates a structured template in Overleaf from scratch.
0:50 – 1:15	Equations, Figures, Tables AMS math environments, including graphics, tabular environment, captions and labels	Hands-on: replicate one equation and one table from a sample paper.
1:15 – 1:40	Principles of Scientific Writing Active vs. passive voice, hedging language, conciseness, precision, avoiding jargon overload	Edit-and-improve exercise: students rewrite a poorly written scientific paragraph.
1:40 – 2:00	Common Writing Mistakes Vague claims, overclaiming results, missing units, inconsistent notation, redundant phrasing	Self-audit checklist provided; students apply it to a short writing sample.

Level-Specific Notes

- Bachelor: Set up an Overleaf document with a title page, sections, one equation, and one figure.

- Master: Use BibTeX integration with Zotero; prepare a document using a journal-specific LaTeX template.
- PhD: Work with multi-file LaTeX projects; explore version control with Git and Overleaf's history feature; prepare using APS or IOP journal templates.

Field Spotlight (30 Min): Material Science | Presenter: Dharendra

- Scope of material science: nanomaterials, semiconductors, polymers, composites, energy materials, and superconductors.
- Characterization techniques: X-ray diffraction (XRD), electron microscopy (SEM/TEM), spectroscopy, and mechanical testing.
- Emerging frontiers: 2D materials (graphene, MoS₂), perovskite solar cells, topological materials, and metamaterials.
- Research-to-product pathways: how material science bridges fundamental physics and industrial application.

Day 6 : Research Proposal Writing & Presentation Skills | 2 Hours

Learning Objectives

- Construct a well-structured research proposal covering all standard sections.
- Formulate a clear, testable, and feasible research question.
- Design and deliver a compelling 8–10 minute research presentation.

Session Outline

Time	Topic	Activity / Notes
0:00 – 0:25	Proposal Structure Title, Abstract, Background, Research Problem, Objectives, Methodology, Expected Outcomes, Timeline, References	Annotated template and a model proposal distributed and walked through.
0:25 – 0:50	Formulating a Research Question PICO and FINER frameworks; specificity, measurability, feasibility, novelty	Students draft 3 candidate research questions; peer and facilitator feedback.
0:50 – 1:15	Scope and Feasibility Time constraints, data availability, computational and lab resources, mentorship alignment	Group discussion: what makes a proposal realistically achievable in 3 months?
1:15 – 1:40	Slide Design for Research Visual hierarchy, use of figures and data, text density, font and color guidelines	Contrast exercise: strong vs. weak slide decks, class identifies differences.
1:40 – 2:00	Presentation Delivery Pacing, clarity, handling Q&A gracefully, managing 8–10 minute time window	30-second elevator pitch exercise: every student presents their idea in 30 seconds.

Level-Specific Notes

- Bachelor: Write a clear one-page proposal with a defined question, simple methodology, and a basic timeline.
- Master: Develop a full three-page proposal with literature context, justified methodology, and measurable objectives.
- PhD: Address contribution to knowledge, originality claims, preliminary data or theoretical framework, and potential publication venues.

Field Spotlight (30 Min) : Econophysics | Presenter: Prasant

- Econophysics: applying statistical mechanics, complex systems theory, and network science to economic and financial phenomena.
- Key topics: power-law distributions in markets, agent-based models, financial contagion, wealth distribution dynamics.
- Methods: time-series analysis, stochastic differential equations, network graph analysis, and Monte Carlo simulations.
- Research entry points: open financial market datasets, interdisciplinary collaboration, and simulation-based research projects.

Week 3 : Individual Proposal Presentations and Group Formation

Week 3 is the culmination of Phase 1. Each student delivers an 8–10 minute presentation of their individual research proposal to mentors, coordinators, and peers. Following all presentations, the coordination team reviews proposals and assigns students to research groups based on shared interest.

Presentation Guidelines

Element	Details
Duration	8–10 minutes per student
Q&A and Feedback	3–5 minutes per student, led by assigned mentors
Required Slides	6–10 slides: Title, Background, Research Problem, Objectives, Proposed Methodology, Expected Outcomes, Timeline
Audience	All NSRF participants, mentors, and NSSR coordinators
Evaluation Focus	Clarity of research question, originality, feasibility, slide quality, and communication effectiveness
Submission Before	Slide file submitted to coordinators at least 24 hours before presentation day

Post-Presentation: Research Group Formation

After all presentations, mentors and coordinators review each student's proposal and declared area of interest. Students are grouped into research teams of three to five members based on thematic alignment, and each group is assigned a dedicated faculty mentor. Group assignments are communicated via official email within 24–48 hours of the final presentation.

- Groups are formed based on alignment of research interest and proposal focus.
- Each group typically includes a mix of academic levels (Bachelor, Master, PhD) to foster peer mentoring where possible.
- Group composition is determined by coordinators in consultation with mentors to ensure research viability and mentor capacity.
- Students who have strong interdisciplinary interests may be placed in mixed-field groups with mentor approval.

3. Group Formation and Official Email Communication

Upon completion of all individual proposal presentations in Week 3, the NSRF coordination team reviews every submission and forms research groups based on field alignment. The email template below is the official communication to be sent to each student confirming their research group, assigned mentor, and first meeting details.

Group Assignment Email Template

TO: [Student Email Address]
FROM: nsrf@nssr.org
SUBJECT: NSRF – Your Research Group Assignment
DATE: [Date of Dispatch]

Dear [Student Name],

Congratulations on completing your individual proposal presentation as part of the NSSR Summer Research Fellowship (NSRF). We were pleased with the quality of ideas and the communication demonstrated by all fellows.

Based on your declared research interests and proposal focus, we are pleased to confirm your research group assignment for the next phase of the fellowship:

Field	Details
Group Identifier	[e.g., NSRF-G04]
Research Area	[e.g., Astrophysics — Galactic Structure and Dark Matter]
Group Members	[Name 1], [Name 2], [Name 3], [Name 4]
Assigned Mentor	[Mentor Name], [Designation], [mentor@nssr.org]
First Group Meeting	[Date] at [Time] — [Venue or Video Link]
Confirmation Deadline	Please reply to this email to confirm by [Date].

Your assigned mentor will guide your group through topic finalization, collaborative literature review, research implementation, and manuscript preparation over the coming weeks. Please use the mentor contact details above to introduce yourself before your first group meeting.

Should you have any questions or concerns regarding this assignment, please contact the NSRF coordination office at nsrf@nssr.org. We look forward to supporting your research journey.

Warm regards,

NSRF Program Coordination Team

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Group Formation Criteria

- Research interest alignment: Students whose proposals share thematic or methodological overlap are placed in the same group.
- Academic level balance: Groups ideally include a mix of Bachelor, Master, and PhD students to encourage peer mentoring and distributed expertise.
- Mentor expertise match: Each group's topic is matched to a mentor whose research background best supports that area.
- Group size: Three to five students per group, adjusted based on total enrollment and mentor availability.
- Interdisciplinary groups: Students with cross-field interests may be accommodated in a dedicated mixed-field group with coordinator approval.

4. Phase 2: Group Research Development (Weeks 4–5)

Phase 2 transitions students from individual preparation into collaborative research. Each group works with their assigned mentor to finalize a research topic, develop a group proposal, and receive structured training in academic research paper writing. The outputs of this phase directly inform the research implementation that follows.

Week 4 : Group Topic Finalization and Collaborative Proposal

Component	Details
Topic Finalization	Each group meets with their mentor to align on a specific, feasible research topic. The selected topic must have a well-defined research question, accessible methodology, and achievable outcomes within the fellowship timeline.
Literature Review	Conduct a focused group literature review identifying 15–25 key references. Sub-topics are assigned to individual members for parallel review, then consolidated.
Role Assignment	Distribute research responsibilities: lead writer, data or simulation lead, figure preparation, and reference manager. Roles are recorded in a shared group charter document.
Collaborative Proposal	Groups produce a joint research proposal (3–5 pages) building on individual proposals, incorporating the group literature review, refined research question, and detailed methodology.
Deliverable	15-minute group proposal presentation to all mentors and coordinators at the close of Week 4. Presentation includes: research question, literature context, methodology, and timeline.

Week 5 : Research Article Writing Training

A dedicated training session equips groups with the skills needed to document and publish their research findings in a format meeting academic standards.

Topic	Content
Paper Structure	Full walkthrough of each section: Abstract (objective, methods, key result, and conclusion in ~250 words), Introduction (funnel structure), Methodology (reproducibility standard), Results (figures-first approach), Discussion (link to literature, limitations, future directions), Conclusion.
Literature Review Writing	Synthesizing sources thematically rather than summarizing paper by paper; building a coherent narrative that establishes the context and justification for the research problem.

Citation Integration	In-text citation techniques, citation density norms, citing figures and datasets, correct handling of paraphrase vs. direct quotation.
Common Pitfalls	Circular reasoning, unsupported claims, overlong methods sections, missing limitations discussion, inconsistent notation, and redundant phrasing.
Manuscript Conversion	How to convert a fellowship research report into a journal manuscript: selecting a target journal, reading author guidelines, and applying a journal-specific LaTeX template.
Deliverable	Each group submits a structured paper outline: section headings, key points per section, planned figures and tables, and a complete reference list.

5. Phase 3: Research Implementation (Weeks 6–11)

Phase 3 is the core research execution period. Over six weeks, each group conducts their approved research under mentor supervision, maintaining consistent progress through weekly reports and short presentations. This phase develops independent scientific thinking, time management, teamwork, and the ability to adapt as findings emerge.

Weekly Structure : Weeks 6 to 11

Component	Description
Literature and Theory	Continue developing the theoretical background. Update the literature review as new relevant publications are identified; document all additions in the shared Zotero group library.
Data Collection or Simulation	Execute data collection, experimental work, computational simulation, or theoretical derivation as defined in the group's approved methodology.
Analysis and Interpretation	Process results, prepare publication-quality figures and tables, and interpret findings in the context of the existing literature.
Group Collaboration Meetings	Minimum one group meeting per week (30–60 minutes), with the mentor present for at least every alternate meeting, in person or via video call.
Weekly Report	A 1–2 page structured report submitted by each group every Friday: work completed, results obtained, challenges encountered, and plans for the following week.
Progress Presentation	5–10 minute group progress presentation each week. Presenting duty rotates among group members to develop each student's communication skills.

Research Quality Standards

- All research must comply with NSSR's academic integrity and responsible research conduct guidelines.
- Research data, code, and working documents must be maintained in a shared, version-controlled repository (e.g., shared Google Drive folder or GitHub repository).
- Any change in research scope or methodology must be communicated to and approved by the assigned mentor before implementation.
- Groups are encouraged to iterate: initial hypotheses may be revised in response to evidence gathered during implementation, with mentor guidance.

Mentor Support Schedule

Each mentor commits to the following minimum engagement schedule throughout Phase 3:

Engagement Type	Minimum Frequency
Group Research Meeting	At least once per week; mentor present in person or via video call for at least every alternate meeting.
Report Feedback	Written or verbal feedback on weekly reports within three days of submission.
Manuscript Review	Full review of draft paper sections as submitted (at minimum once during Weeks 9–10).
Pre-Symposium Rehearsal	Review and rehearsal of the final presentation in Week 11, with written feedback on slides.

6. Phase 4: Final Research Symposium (Week 12)

The Final Research Symposium is the flagship conclusion of the NSRF program. All groups present their research findings in a formal academic setting open to NSSR faculty, mentors, invited guests, and institutional partners. Outstanding work is recognized and pathways toward publication or extended research are discussed with each group.

Symposium Day Agenda

Agenda Item	Description
Opening Session	Welcome address by the NSSR Director. Overview of the NSRF program, its goals, and the 12-week journey completed by all fellows.
Group Presentations I	The first half of groups present their research findings. Format: 10–15 minutes per group, covering background, methodology, results, and conclusions.
Q&A Panel I	Mentors, faculty, and invited experts direct questions to presenting groups. Five minutes of Q&A per group.
Group Presentations II	Remaining groups present their findings in the same format.
Q&A Panel II	Open floor Q&A for the second presentation block.
Research Highlights	Mentors share brief observations on notable findings, interesting methodologies, and student growth across groups.
Awards and Certificates	Recognition of outstanding projects, distribution of NSRF Fellowship Certificates, and publication pathway announcements.
Closing Remarks	Closing address, networking session, and guidance on post-fellowship opportunities including NSSR's extended research program.

Final Submission Requirements

Each group must submit all materials listed below before symposium day to be eligible for presentation and certification:

Document	Specification
Final Research Report	8–15 pages formatted in IEEE or APA style. All standard sections required: Abstract, Introduction, Methodology, Results, Discussion, Conclusion, and References.
Presentation Slides	10–15 slides submitted to coordinators at least 48 hours before symposium day for technical review and formatting check.
Supplementary Materials	Data files, analysis code, or simulation files (where applicable) uploaded to the shared NSSR repository before symposium day.

Publication and Continuation Pathway

- Groups producing original and sufficiently developed work will be formally recommended for submission to a peer-reviewed journal or academic conference.
- NSRF coordinators and mentors will provide guidance on manuscript finalization, journal selection, and cover letter preparation for qualifying groups.
- Students wishing to extend their research beyond the fellowship period may apply for NSSR's extended research program, subject to mentor availability and project viability.

7. Expected Outcomes

Upon successful completion of the full 12-week NSRF program, each fellow is expected to demonstrate the following achievements:

Outcome	Description
NSRF Fellowship Certificate	Official certificate issued by NSSR recognizing full 12-week program completion.
Proposal Writing Experience	Both individual and group proposals written, presented, and refined through structured mentor feedback.
Scientific Communication	Demonstrated skills in oral presentation, academic writing, LaTeX typesetting, and slide design.
Research Tools Proficiency	Practical competence in Zotero, Overleaf/LaTeX, academic databases, and research data management.
Collaborative Research	Firsthand experience of team-based research with defined roles, shared goals, and mentor-supported workflows.
Final Research Paper	A complete research paper or detailed publication-ready manuscript draft meeting academic standards.
Publication Opportunity	Selected projects formally recommended for peer-reviewed journal or conference submission.
Field Exposure	Awareness of six diverse research fields through structured spotlights: particle physics, astrophysics, plasma physics, biophysics, material science, and econophysics.

8. Quick Reference: Complete 12-Week Timeline

Wk	Phase	Activities	Deliverable
1	Foundation	Research orientation, scientific method, literature databases, reading scientific papers	Paper reflection submission
2	Foundation	Zotero, LaTeX and Overleaf, proposal writing, presentation skills training	Individual research proposal (draft)
3	Foundation	Individual proposal presentations (8–10 min per student) with Q&A and mentor feedback	Proposal presentation; group assignment email sent
4	Group Dev.	Group meeting, topic finalization, literature review, role assignment, collaborative proposal	15-minute group proposal presentation
5	Group Dev.	Research paper writing training: structure, literature review, citation integration, manuscript preparation	Draft paper outline with section plan and reference list
6	Research	Literature review, theoretical framework development, initial data or simulation work	Weekly report + 5–10 min progress presentation
7	Research	Data collection or simulation runs, analysis setup, initial findings	Weekly report + progress presentation
8	Research	Continued data collection, preliminary result analysis, figure preparation	Weekly report + progress presentation
9	Research	Result interpretation, draft Introduction and Methodology sections of the research paper	Weekly report + progress presentation
10	Research	Draft Results and Discussion sections; mentor manuscript review; revisions	Weekly report + progress presentation
11	Research	Complete final manuscript draft, rehearse symposium presentation, prepare slides	Final report draft + symposium presentation slides
12	Symposium	Final research presentations, panel Q&A, awards, certificates, publication pathway discussion	Final report, slides, NSRF Certificate, publication recommendation

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